

VIGILANCE

Predictive Cash-Out Intelligence for Cybercrime Complaints

SIH 2026 STRICT TECHNICAL EVALUATION

Problem Statement SIH26184 — Team UNFAZED

Evaluated against the actual codebase (d:\skcet\SIH-PredicTrace), the real evaluation_report.json,
and the actual submitted file `SIH2026 PPT.pptx` — all 7 slides and 5 embedded graphics
inspected directly, not assumed from memory.

Prepared as an independent technical jury simulation.

1 Overall SIH Evaluation

Category	Score	Why
Problem Understanding & Relevance	13/15	Directly targets the exact PS ask ("forecast likely cash withdrawal locations"). Loses marks because two of the PS's four named deliverables — Risk Heatmap Dashboard (GIS heatmap with time/location/crime-category drill-down) and Alert & Notification System (SMS/email/API to LEAs/banks/I4C) — are not built and not honestly flagged as gaps anywhere in the deck.
Innovation / Novelty	15/20	Point-in-time graph discipline and lift-based urgency framing are genuinely above hackathon-median rigor. Loses marks because the core method (XGBoost + hand-engineered graph features + Louvain) is well-established ML engineering, not a new technique — the novelty is in assembly and discipline, not invention.
Technical Depth	16/20	Real point-in-time graph construction, real calibration, real SHAP on the raw booster, real multi-seed stability check. Loses marks: entity extraction is 3-field regex, not the NLP claimed on slide 3; no automated retraining despite the claim; single-city-scale synthetic data (200 candidates) undermines depth-at-scale.
Feasibility & Viability	11/15	Working prototype is real feasibility evidence. Loses marks: "already supports retraining" is false — manual CLI script, zero automated trigger. "Production-ready architecture" overstates a codebase with no auth, no rate limiting, no migrations, CORS wide open.
Impact & Benefits	10/15	Real, computed numbers used rather than invented ones — commendable. Loses marks: the 4.4-hour figure is presented as a system benefit when it's a property of synthetic dataset timestamps unrelated to the model; "national-scale impact... crores" has no computation shown.
Prototype / Implementation	9/10	Strongest section of the submission — genuinely running, multi-page, real DB writes, real decision logging. One point off because PS-named deliverables (heatmap, alerting) are absent from what's actually running.
Presentation & Communication	2/5	Clean template use, good real charts. Slide 6 (Research and References) is the unedited template placeholder text — a required slide submitted blank.

TOTAL SCORE **Competitive but Needs Improvement**

76 / 100

Strong enough prototype and enough real engineering discipline to survive a technical panel's questioning — but an unedited blank slide and an internally contradictory headline number (see §10) are exactly the kind of unforced errors that knock a genuinely good project out of a competitive round before the judges even get to the technical merits.

2 Codebase Credibility Check

PPT Claim	Actual Implementation	Status	Judge Risk
Point-in-time graph, no future leakage	Confirmed: graph rebuilt per-complaint, <code>as_of</code> strictly < <code>filed_at</code>	IMPLEMENTED	Low
Fraud Network Discovery / Cross-Complaint linking	Louvain community detection on projected account graph, no fuzzy/entity-resolution layer	IMPLEMENTED (narrower than it sounds)	Medium — no identity-matching across victims by name/phone
Predicts Top 5 cash-out locations	Real ranked XGBoost output over candidate withdrawal points	IMPLEMENTED	Low
"200 ATMs" (slides 4, 5)	Confirmed: <code>withdrawal_points.csv</code> = 200 rows	IMPLEMENTED	Low
"6,000–8,000+ Possible ATMs" (slide 2 graphic)	Actual validated candidate set = 200, everywhere else in the same deck	OVERSTATED / SELF-CONTRADICTED	SEVERE — see §10
Explainable & Calibrated Predictions	Real SHAP on raw booster; real reliability table (2 bins, n=5,304 / 216)	IMPLEMENTED	Low-Medium — only 2 bins have enough volume
"48.6% Precision@5 vs 12.3% Baseline"	Confirmed exact match to <code>evaluation_report.json</code> (single seed)	IMPLEMENTED, reproducible	Low if seed variance disclosed
"4.4-hour lead time"	Real number, but measures <code>withdrawn_at</code> – <code>filed_at</code> on synthetic timestamps — independent of the model or ATM count	OVERSTATED as a system benefit	High
"Entity Extraction: accounts, UTRs, timestamps, locations, beneficiaries"	<code>entity_extraction.py</code> : regex over 3 fields only — <code>bank_name</code> , <code>ifsc_code</code> , <code>amount</code> . No UTR/timestamp/location/beneficiary extraction exists.	OVERSTATED	High — disprovable in 30 seconds
"Continuous retraining: already supports retraining"	<code>ml/train.py</code> is a manual <code>argparse</code> CLI. Zero scheduler, zero trigger from /decision endpoint.	OVERSTATED / FALSE AS WORDED	High
"Production-ready architecture: SQLite → PostgreSQL"	<code>psycpg2</code> -binary present but never connected; no Alembic/migrations; SQLAlchemy makes the swap plausible, not proven	ARCHITECTURAL	Medium
Human-in-the-loop / audit trail	Real DB status write, but <code>event_log</code> is a 200-item in-memory deque; complaint status force-reset to "open" on every backend restart	PARTIAL	Medium — "is this decision permanent" is more complicated than implied
"Integrated with NCRP Workflows"	No NCRP API, format, or schema mapping exists anywhere	ARCHITECTURAL / ASPIRATIONAL	Medium
Risk Heatmap Dashboard (PS deliverable)	Discrete markers/circles by rank+confidence, not a density heatmap; no time/location/crime-category filters	NOT BUILT (not claimed either)	Medium
Alert & Notification System (PS deliverable)	Zero SMS/email/webhook/API dispatch code anywhere	NOT BUILT (not claimed either)	Medium

3 Technical Judge Attack

Q1. Slide 2 says 6,000–8,000 ATMs, slide 5 says 200. Which is it, and which number is your Precision@5 actually measured against?

Why asked: five-second cross-reference any judge glancing between two slides will catch. · Difficulty: 3/10

Ideal answer: "200 — that's what's validated. 6,000–8,000 was an illustrative real-world estimate that shouldn't have been in the deck without the caveat that we haven't tested at that scale."

Evidence: evaluation_report.json n_test_complaints=138 over a 200-candidate set; withdrawal_points.csv = 200 rows.

Weakness: no answer avoids admitting the 6-8k number was an error — only mitigation is owning it immediately.

Q2. Your entity extraction slide lists UTRs, timestamps, locations, beneficiaries. Show me one UTR you extracted.

Why asked: judges probe the one line item they suspect is padding. · Difficulty: 4/10

Ideal answer: "Today it's 3 fields — bank, IFSC, amount — via regex against templated narratives. UTR/location/beneficiary extraction is scoped but not built; the diagram overstates current state."

Evidence: nlp/entity_extraction.py:15-35.

Weakness: none if answered honestly; becomes a credibility collapse if defended as-is.

Q3. Why gradient-boosted trees and hand-engineered graph features instead of a GNN?

Why asked: standard for anyone doing "graph + ML." · Difficulty: 6/10

Ideal answer: "At this data volume (~700 complaints, ~1,700 accounts) a GNN would overfit and add an unexplainable embedding layer — SHAP over engineered features gives per-prediction explainability a GNN embedding can't."

Evidence: ADVANCED_FEATURE_COLUMNS in features.py (11 named features); SHAP integration in explain.py.

Weakness: legitimate only because explainability was prioritized — must not claim a GNN "wouldn't work," only that it wasn't the right tradeoff.

Q4. How is the target variable defined — what exactly is the model predicting?

Difficulty: 5/10

Ideal answer: Binary "is this the true cash-out withdrawal point for this complaint," ranked across all candidates by predicted probability, using the real synthetic ground-truth withdrawal event per complaint.

Evidence: ml/train.py, features.py candidate generation, ground-truth withdrawal_events table.

Weakness: ground truth only exists because the generator wrote it in — a synthetic-label problem, not observed real-world labeling.

Q5. Your false-positive rate at high confidence is 72.5% per your own eval file. Why isn't that on a slide?

Why asked: a judge who opens evaluation_report.json will find "false_positive_rate": 0.725. · Difficulty: 8/10

Ideal answer: "HIGH confidence means 5x lift over random chance in a 200-way ranking, not >50% absolute confidence — no model can hit 50% absolute in a 200-way problem. This is disclosed, not hidden; it's why every prediction requires human sign-off."

Evidence: HIGH_CONFIDENCE_THRESHOLD = 0.125, lift-based urgency framing.

Weakness: correct but sounds evasive unless delivered with total command — rehearse verbatim.

Q6. Is your baseline fair, or a strawman?

Difficulty: 6/10

Ideal answer: "Same train/test split, same target, same evaluation code — only the feature set differs. Not tuned as hard as the advanced model, which we disclose."

Evidence: evaluate.py computing both models through identical precision_at_k.

Weakness: baseline tuning effort vs. advanced model's tuning effort is undocumented.

Q7. What happens when the model's top-5 is wrong?

Difficulty: 4/10

Ideal answer: Nothing automated — investigator sees SHAP reasons, a near-tie warning when top-2 confidences are within 3 points, must sign off. A wrong top-5 wastes time; it does not trigger autonomous action.

Evidence: isCloseCall warning in CaseWorkspace.jsx; /decision requiring explicit human input.

Weakness: no measurement of how often investigators would actually catch a wrong shortlist — asserted, not measured, safety.

Q8. What's your retraining story under concept drift?

Difficulty: 7/10

Ideal answer (corrected): "Retraining today is a manual script over the full historical set; automatic feedback from investigator decisions into retraining is designed-for but not built."

Evidence: ml/train.py CLI-only.

Weakness: the deck currently claims this is "already" supported — must be walked back before this question is asked.

Q9. How does this scale from 200 ATMs / 5 cities to a real NCRP feed of ~8,000 complaints/day nationally?

Difficulty: 7/10

Ideal answer: Graph construction and Louvain are most likely to need re-architecture at national scale; XGBoost inference itself is cheap and scales fine.

Evidence: none in the repo — honestly unaddressed.

Weakness: real, unmitigated gap.

Q10. Real bank data is noisy, has missing fields, delayed reporting. Does your pipeline survive that?

Difficulty: 6/10

Ideal answer: Not validated — the synthetic generator produces clean, complete records by construction; real STR data will have missing counterparties, delays, and format inconsistencies never tested against.

Evidence: none — an honest gap.

4 Synthetic Data & Evaluation Audit

- **Ground truth:** real, generator-assigned withdrawal events per complaint — internally consistent, not label noise.
- **Temporal split:** real and enforced (as_of strictly-less-than filtering) — the one thing that would genuinely impress a judge who checks for it.
- **Scale:** 690 complaints, 1,667 accounts, 2,468 transfers, 200 withdrawal points, 5 cities, 25 ground-truth rings — realistic relative structure, small absolute volume (138 test complaints); mitigated by 5-seed stability ($P@5 = 46.2\% \pm 3.5\%$).
- **Noise:** clean, template-based narratives and complete transaction chains — real NCRP/bank data will have missing fields, delayed reporting, inconsistent formats. Not modeled at all.
- **Geographic distribution:** real, named, verified-inland localities per city (a real coastal-prediction bug was found and fixed this session — evidence the team actually looked).

Verdict: 6/10. Convincing that the methodology (point-in-time discipline, calibration, multi-seed stability) is sound. NOT convincing that the absolute numbers (48.6%, 4.4h) would hold on real, noisy, national-scale data.

Claims to make: "Point-in-time discipline prevents a real leakage class most fraud-ML has." "Multi-seed stability shows this isn't a lucky split." "Synthetic-but-structurally-realistic, disclosed as such."

Claims to NOT make: "48.6% precision in the real world." "4.4-hour lead time for investigators." "Validated at scale." "Production-ready."

5 Metrics Audit

Metric	Reproducible?	Correct calc?	Fair comparison?	Judge-challengeable?
Precision@1/@5 (4.3/12.3 vs 17.4/48.6)	Yes — matches evaluation_report.json exactly	Yes	Reasonably — same split/code, feature sets differ (disclosed)	Yes, on baseline tuning effort
Multi-seed stability (46.2%±3.5%)	Yes	Yes	N/A	Low risk — strengthens the claim
Calibration (2 bins)	Yes	Yes	N/A	Yes — only 2 bins have volume; say so unprompted
Lead time (4.4h)	Yes, mechanically	Yes, as computed	No — not comparing anything, a raw timestamp gap presented as a system output	High — must be reframed
False-positive rate (72.5%, unshown)	Yes	Yes	N/A — not shown, but exists in the artifact a judge could open	High if found

6 Novelty Audit

Genuine technical differentiation

Point-in-time graph reconstruction per complaint (prevents a leakage class most fraud-ML ignores); lift-based urgency (5x over random chance in a 200-way space) instead of a meaningless absolute-confidence threshold; complaint-level (not row-level) calibration split.

Strong engineering integration, not individually novel

XGBoost + SHAP + Louvain + calibration — every piece is standard; the discipline of point-in-time-consistent feature computation across all of them simultaneously is the actual engineering achievement, and it's real (verified this session, including a caught-and-fixed leakage bug).

Standard technology — should not be called innovation

XGBoost itself, Louvain itself, SHAP itself, React/FastAPI/Leaflet — none of this is novel and the deck correctly doesn't claim it is.

Why pick this over a conventional fraud dashboard: because a conventional dashboard scores a transaction; this predicts a place, ranks it, explains it, and is provably free of the temporal leakage that makes most such demos' accuracy numbers meaningless.

7 PPT-by-PPT Review

Slide	Assessment	Class.	Exact fix
1 — Title	Correct PS ID, title verbatim, theme/category/team all correct.	KEEP	—
2 — Idea	Strongest slide — problem/solution/novelties all present with real numbers. Weakest element: the 6,000–8,000 ATM graphic.	MODIFY	Fix the ATM number to 200, or remove that graphic entirely.
3 — Technical Approach	Workflow diagram accurate and honest (7 real steps). Tech stack list accurate and specific. Entity-extraction bullet overstates 4 fields that don't exist.	MODIFY	Change step 2 to "bank, IFSC code, amount" or remove the extra fields.
4 — Feasibility	Text-dense (6+4+4 bullets). "Already supports retraining" is the one factually false line.	MODIFY	Change to "designed to support," not "already supports."
5 — Impact	Lead-time language here is more careful than slide 2's — good, but the two slides now disagree in tone about the same number.	MODIFY	Copy slide 5's careful framing back onto slide 2.
6 — Research and References	Blank template placeholder text. Required slide submitted empty.	REBUILD	Add: Louvain (Blondel et al. 2008), XGBoost (Chen & Guestrin 2016), SHAP (Lundberg & Lee 2017), NCRP/I4C public reporting stats, RBI fraud-reporting circulars.
7 — Instructions	Not a content slide.	DELETE before PDF export	Per its own text.

8 Presentation Quality

- **10-second comprehension:** yes — slide 2 alone communicates problem, solution, and 4 novelties clearly.
- **Working-system credibility:** yes, more than most — real workflow diagram, real tech stack, real numbers all read as "built," not "planned."
- **Numbers memorable:** yes (48.6% vs 12.3%, 40×, top-5) — undermined the moment two slides contradict each other on scale.
- **Narrative logical:** mostly — breaks at the ATM-count inconsistency and the lead-time claim's inconsistent framing across slides 2 and 5.
- **Unnecessary text:** slide 4's three-box layout is dense; could lose 2-3 bullets without losing content.

The 3 slides that matter most: Slide 2 (first impression), Slide 3 (where "is this real" gets judged), Slide 5 (where the numbers get scrutinized). Slide 6 currently actively damages all three by being blank.

9 Demo Evaluation

Strongest possible sequence

1. Command Center, live (0–15s) — real dashboard, live feed moving, real stat cards.
2. Open a case → Fund-Flow graph → Cash-out Prediction ranked map with confidence (15–40s) — the single strongest "wow" moment: a real graph traced and a real ranked shortlist appearing live with SHAP reasons underneath.
3. Click into a ring from that case → Ring Detail page showing 30–40 other linked complaints (40–55s) — second-strongest moment; "one complaint just exposed 38 others" lands harder than any slide.
4. Approve → Intervention Brief download (55–70s) — closes the loop with a tangible artifact.

Skip: Reports/Analytics (too technical for the first minute — hold for Q&A), Settings, Audit Trail (mention verbally).

First 60 seconds must show: a real complaint → a real ranked prediction with a real confidence score and a real reason.

Most damaging possible failure: the activity simulator producing a visibly duplicate/nonsensical Live Feed event while a judge watches (a real bug found and fixed this session — confirm no regression). Second: a live confidence percentage not matching whatever's on a slide — rehearse against the live app, not old screenshots.

10 Red Flags

Claim	Why Risky	Severity	Action	Better Wording
"6,000–8,000+ Possible ATMs" (slide 2)	Directly contradicted by slide 5's "200 ATMs" — same deck, same team	HIGH	Remove/fix now	"200 candidate cash-out points (prototype scale, city-level)"
"4.4 hours lead time" as headline benefit	Measures dataset timestamp gap, not a delivered capability; wrongly implied to connect to ATM count	HIGH	Qualify	"In our modeled data, a real filing-to-cashout window exists — averaging 4.4h — during which real-time scoring is actionable"
Entity Extraction: accounts, UTRs, timestamps, locations, beneficiaries	Trivially disprovable — only 3 fields exist	HIGH	Remove extra fields	"Entity Extraction: bank name, IFSC code, amount (regex-based)"
"Continuous retraining: already supports"	False as worded — manual script only	HIGH	Remove "already"	"Architecture supports periodic retraining on newly closed cases (manual today, automatable)"
"Production-ready architecture"	No auth, no rate limiting, no migrations, wide-open CORS	MEDIUM	Qualify	"Production-ready data layer (SQLAlchemy ORM, Postgres-compatible); auth/hardening out of scope for this prototype"
"Designed for real-world adoption... NCRP Workflows"	No NCRP integration code exists	MEDIUM	Qualify	"Designed to integrate with NCRP data formats (not yet implemented)"
"48.6% Precision@5" without seed disclosure on chart	A judge asking "one seed?" deserves a chart that already answers	LOW	Add footnote	Add "(46.2%±3.5% across 5 seeds)" as chart subtitle
"12.2%" baseline P@5 on chart	Actual value is 12.3%	LOW	Fix	12.3%
"700+ complaints"	Actual = 690	LOW	Fix	"690 complaints"
Slide 6 blank	Required slide, unedited template text	HIGH	Fix before submission	Fill with real citations (§7)

11 Judge's Final Verdict

FINAL SCORE

76 / 100

Top 5 Strengths

1. Point-in-time graph discipline is real, verified in code, and rarer than any other single technical claim in this deck.
2. The prototype is genuinely running end-to-end — DB writes, real decisions, real SHAP, not a mockup.
3. Multi-seed stability check ($46.2\% \pm 3.5\%$) pre-empts the "lucky split" objection most teams never think to address.
4. Real, computed numbers used throughout (evaluation_report.json), not invented statistics.
5. Ring Detail / cross-complaint discovery is a genuinely strong, demoable differentiator.

Top 5 Weaknesses

1. Slide 2's 6,000–8,000 ATM number directly contradicts slide 5's 200 — an unforced, easily-caught internal contradiction.
2. Slide 6 (References) is blank template text.
3. Entity-extraction diagram claims 7 fields; 3 exist.
4. "Continuous retraining already supports" is factually false as worded.
5. Zero implementation of two of the PS's four named deliverables (Risk Heatmap, Alert/Notification System) with no acknowledgment of the gap.

Top 5 Things to Fix Before Judging

1. Fix the ATM number on slide 2 to 200 (or remove that graphic entirely).
2. Fill slide 6 with real references (list in §7).
3. Correct the entity-extraction claim to the 3 real fields.
4. Change "already supports retraining" to "designed to support."
5. Fix $12.2\% \rightarrow 12.3\%$ and $700+ \rightarrow 690$.

Top 10 Questions to Prepare For

See §3 (Q1–Q10) — in priority order, Q1, Q2, Q5, and Q8 are the ones most likely to actually get asked and most damaging if unprepared.

Biggest Technical Risk

No validation at real-world candidate-set scale (200 vs. thousands of real ATMs) — every headline precision number could be materially different at real scale, and this is currently undisclosed.

Biggest Presentation Risk

The 200 vs. 6,000–8,000 self-contradiction — it takes a judge under ten seconds to find and it costs credibility on every other number in the deck by association.

Strongest Differentiator

Provable absence of temporal leakage in a graph-based fraud model — most competing teams doing "AI fraud detection" will not have thought about this, and this team can show the actual fix in code.

Strongest Demo Moment

Opening one complaint and watching it surface 30–40 linked accounts and other victims' cases live, via Ring Detail.

Most Questionable Claim

"6,000–8,000+ Possible ATMs" narrowed to 5 — unvalidated, contradicted by the team's own other slide, and the single easiest thing in this entire submission for a judge to disprove in real time.

Final Verdict

MAYBE

As a real SIH evaluator: the underlying engineering clears the bar for a technical round — point-in-time graph discipline and honest multi-seed evaluation are not things most teams at this level do. But this specific PPT, as it currently stands, contains a self-contradicting headline number and a blank required slide. Neither is a technical failure — both are avoidable editing failures — but in a competitive round evaluators are actively looking for reasons to cut the field down, and a five-second cross-slide contradiction is exactly the kind of thing that ends a submission's run before its actual merits get discussed. Fix the five items above and this moves to a clear YES.

Why should this project win over the other teams, honestly

Not because it's flashier — it isn't, and shouldn't try to be. It should win because it is one of the only submissions in the room that can survive a technical judge actually opening the code: the leakage-prevention claim is real and demonstrable, the calibration claim is real and demonstrable, the multi-seed stability claim is real and demonstrable. Most competing "AI fraud detection" dashboards will collapse under that exact scrutiny because their accuracy numbers were computed carelessly or not at all. This team's evidence is real — it just needs to stop overstating the parts that aren't, because right now the deck's own internal contradiction is doing more damage to its credibility than any question a judge could ask.

Prepared as an independent, evidence-based technical audit — all claims cross-checked against the live codebase, the real evaluation artifact, and the actual submitted PPTX file (text and embedded graphics) rather than assumed. Not an official SIH scoring outcome.